Megh Bahadur KC
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PROFESSIONAL SUMMARY

Ph.D. candidate in Civil Engineering at the University at Buffalo, specializing in sustainable public transport, freight systems and intelligent mobility through optimization, machine learning and deep reinforcement learning. Over 8 years of combined research, academic and industry experience developing efficient optimization tools, predictive models, techno-economic analyses, advanced data analytics and inclusive planning/policies at NREL, NYCDOT, Utah State University and the Department of Roads. Proven achievements include 7+ peer-reviewed publications, 17+ conference presentations, 2 internships, multiple awards and fellowships, effective undergraduate and graduate level teaching and hands-on project management as a Highway Engineer in transport infrastructure design and implementation.

EDUCATION

- **Ph.D.** in Transportation System Engineering**; University at Buffalo, SUNY**- GPA: 4.00, by May 2026

Dissertation Title: "Accelerating Bus Transit Electrification through Optimization, metaheuristics and Reinforcement Learning Frameworks" Advisor: Dr. Ziqi Song, Associate Professor

- **M. Sc.** in Construction Management; Pokhara University, Nepal- GPA: 3.95
- **B. E.** in Civil Engineering; Institute of Engineering, Tribhuvan University, Nepal; ranked top 4

RESEARCH INTERESTS

- **Operation Research:** Optimization, Application of operation research on electrified transportation systems, Reinforcement Learning for transport system efficiency, Multimodal Freight modeling
- AI/RL/ML in transportation, ITS and advanced transportation applications (V2X, CAV), Digital twins
- Travel Demand Modeling, Travel Behavior & Discrete Choice Modeling
- Inclusive Planning & Policy for urban land use & transportation system analysis

WORK EXPERIENCE

SN.	From	To	Employer	Position	Supervisor
7	01- 2025	to date	University at Buffalo, SUNY	Graduate Research Assistant (GRA)	Asso. Prof. Dr. Ziqi Song
6	09-2025	to date	New York City Department of Transportation	Freight Analytics Intern	Catherine Ponte, Team lead
5	05-2024	01-2025	National Renewable Energy Laboratory, Denver, CO	Transport System Analysis Intern	Jason Lustbador, Group Manager
4	08- 2021	01-2025	Utah State University	GRA	Asso. Prof. Dr. Ziqi Song
3	03- 2020	08- 2021	Pokhara University, Nepal	Asst. Professor	Dr. KN Pandey, Principal
2	07- 2015	08- 2021	Ministry of Physical Infrastructure and Transport, Nepal	Highway Engineer	Director General, Department of Roads
1	03- 2014	02- 2015	Tribhuvan University, Nepal	Assistant Lecturer	Prof. Dr. Rajan Suwal

1) Graduate Research Assistant @ University at Buffalo

- Ongoing PhD Dissertation research chapters:
  - *Multi-Agent Deep Reinforcement Learning for Integrated Bus Scheduling and Charging Management Under Uncertainty in Multi-Line Public Transit Operation.*
  - *Deep Reinforcement Learning Framework for Simultaneous Fleet Optimization and Charging for Multi-line Bus Transit System in a Dynamic Environment.*
  - *Integrated Optimized Charging Decision and Optimal Battery Sizing of V2G Enabled Bus System Including Demand Charges & Battery Degradation.*
  - *Bus Electrification: Fleet Optimization and Charging Management Considering Partial Recharges, TOU and Demand Charge.*

- Led "Residential Location Choice Modeling" project for WFRC Utah, identified key policy intervention for TOD & pedestrian-friendly infrastructure, contributing to an increase in transit ridership
- Developed and published freight-analytics-dashboard Python package on PyPI

Grant & Proposal Writing Contribution

- NYSERDA RFP 5965- *Electric School Bus Transition Support Contractor*, $1M, 2025-2028
  - Drafted Statement of Work (SoW) section [submitted]
- DOE, Joint Office of Energy and Transportation (JOET), Concept paper in response to FOA – *Expanding E-Mobility Solutions through Electrified School Buses*, $250,000.
  - Drafted Background and Proposed Methodology section

2) Freight Analytics Intern @ New York City Department of Transportation (NYCDOT)

- Designing a VMT assessment and GHG reduction tool for NYC freight transportation
- Quantifying the effect of freight policy intervention such as NY congestion pricing, off-hour delivery programs in terms of VMT, GHGs and air pollution
- Complex public and private dataset analysis including StreetLight, INRIX, Geotab, ATRI

3) Transport System Analysis Intern @ National Renewal Energy Laboratory (NREL)

- Developed & coded Freight Analysis Framework (FAF) & CFS flow containerization model & calibration
- Robust optimization of intermodal freight operations in disruptive events (national-scale) through modeling & simulation of containerized freight flow
- Built last-mile delivery vehicle routing: VRP, CVRP, VRPPD, and benchmarked with Google OR-Tools
- Contributed on SMART 2.0 Freight project by successfully developing & integrating the electric variant of NREL's open-sourced Freight Integrated Simulation Model (FRISM) for heterogeneous mixed fleet combination using VNS metaheuristics
- Formulated methodological framework for the county-level freight flow disaggregation model algorithm
- Created & hosted an interactive Container Freight data visualization Dashboard in Streamlit
- Received Excellent (best) performance grade during review for the year from the Group Manager

4) Graduate Research Assistant @ Utah State University

- Developed and linearized a multi-objective optimization model in GAMS and produced a hybrid metaheuristic algorithm in Python 3.10 with solver efficiency (730 seconds) in large-scale networks
- Produced and enhanced a multistage multi-objective linear /nonlinear optimization model with the achievement of efficient recharging and scheduling of EVs in 1,000+ node networks in 2 stages
- Automated a space-time prism (STP) modeling by developing an ArcGIS Pro algorithm in Python using the arcpy environment to measure accessibility for people with disabilities in only 1.5 months
- Performed techno-economic analysis on "Integrated Optimized Charging Decision and Optimal Battery Sizing of V2G Enabled Electric School Bus System Including Demand Charges & Battery Ageing."
- Led the best group project in machine learning titled "Fake News Detection: Investigating ChatGPT" by using Naive Bayes, Random Forest, and Neural Networks among 19 groups
- Conducted research on Discrete Mode Choice Modeling using Nested and Multinomial Logit in Pandas Biogeme, resulting in policy recommendations for the Inclusive Transportation System in Utah
- Completed term project on Change Detection Analysis for Selected Municipalities in Utah using ArcGIS Pro
- Conducted emission modeling for a new electrified metro line system, estimating mode shift, which reduced 6% annual GHG
- Participated in Kaggle Competition as a part of Final Project on Machine Learning and Data Analytics

5) Highway Engineer/ Site Project Manager

- Completed feasibility studies for multi-million-dollar Highway Projects, cost-benefit analysis, specifications.
- Obtained design approval for a 17-mile 6-lane road project, representing the Government of Nepal.
- Managed the procurement process for a massive ~$40M road project, including bid document preparation, contract work specification, bid publication, evaluation, and contract award.

- Served as a site project manager, leading a team of 3 engineers, 8 sub-engineers, and 15 site supervisors.
- Skillfully coordinated with the communication manager for the upper body during the project's execution.
- Recognized with the Best Employee Award in 2019 for reducing office annual costs by 19% through optimizing employee schedules and operations.
- Part-time Assistant Professor, lectured to first-year MSc students on "Project Planning and Control."

RESEARCH EXPERIENCE

Peer-Reviewed Journal Articles

- **KC, M. B.,** & Song, Z. (2025). Evolving School Transportation: Integrated Dynamic Route Optimization and Partial Charging for Mixed Fleets. In arXiv e-prints. https://doi.org/10.48550/arXiv.2511.00600
- **KC, M. B.** (2025). Land Use Transition and City Desirability: Case Study of Cache County Over Two Decades. *Growth and Change, A Journal of Urban and Regional Policy, 2025*; 56(4). DOI: https://doi.org/10.1111/grow.70057
- **KC, M.B.,** Song, Z., Park, K. & Christensen, K. (2025). *Understanding Mode Choice Behavior of People with Disabilities: A Case Study in Utah*. In arXiv e-prints. https://doi.org/10.48550/arXiv.2511.11748
- **KC, M.B.,** Mishra, A.K. & Karki, R. (2020). Perception Based Assessment of Bidding Practices and Effectiveness of E-Bidding Systemin Nepal. *East African Scholars of Economics, Business and Management* 2020; 3(7): 588-597. Link
- Mishra, A.K., **KC, M. B.** & Aithal, P.S. (2020). Association of Number of Bidders and Minimum Bid Ratio (AEr) with Effect of E-bidding of Different Project. *International Journal of Management, Technology, and Social Sciences* 2020; 5(2): 201-215. Link
- **KC, M. B.,** & Mishra K. (2019). Bidding Trends of Contracts based on Types and Sizes of Projects under Road Divisions Butwal and Shivapur. *Journal of Advanced Research in Construction and Urban Architecture* 2019; 4(4): 7-16. Link

Under Review and Under-Preparation Journal Papers

- **KC, M. B.,** & Song, Z. (2026). Multi-Agent Deep Reinforcement Learning for Integrated Bus Scheduling and Charging Management Under Uncertainty in Multi-Line Public Transit Operation. *[work in progress].*
- **KC, M. B.,** Song, Z. & Chaoyue Zhao. (2026). Deep Reinforcement Learning Framework for Integrated Fleet and Charging Management for Electric Bus Transit System in Dynamic Environment. *[TRBAM 2026].*
- **KC, M. B.,** & Song, Z. (2026). Bus Electrification: Fleet Optimization and Charging Management Considering Partial Recharges, TOU and Demand Charge. *[TRBAM 2026].*
- **KC, M. B.,** & Song, Z. (2025). Integrated Optimized Charging Decision and Optimal Battery Sizing of V2G Enabled Electric School Bus System Including Demand Charges & Battery Ageing. *[TRB 2024, under review, Applied Energy].*
- **KC, M. B.,** & Song, Z. (2025). School Transport Electrification - Adoption, Methods, Policy and Strategies: A Comprehensive Review. *[Under review, Transport Reviews].*
- **KC, M. B.,** & Song, Z. (2025). Multi-Stage, Multi-School Electric Bus Routing and Scheduling Optimization with recharging schedule and re-use of Buses using metaheuristics. *[TRB 2024 & under review, TR Part D].*
- **KC, M. B.,** Song, Z., Park, K., Chamberlin, B. & Christensen, K. (2025). Household Residential Location Choice Modeling for People with Disabilities in Utah. *[TRBAM 2026 & under review, TR Part A.*

RESEARCH PROJECTS

- INtermodal Freight Optimization for a Resilient Mobility Energy System (**INFORMES**), Advanced Research Projects Agency-Energy's (ARPA-E), US **Department of Energy (DOE)** @ National Renewal Energy Laboratory, Denver, Co, USA
- Route Optimization, On-route Charging and V2G Integration on Transportation thrust for **ASPIRE** (Advancing Sustainability through Powered Infrastructure for Roadway Electrification) National Science Foundation **(NSF) Engineering Research Center**, Logan, Utah
- Increasing Affordability, Energy Efficiency, and Ridership of Transit Bus Systems through Large-Scale Electrification, US Department of Energy (DOE)

- Systems and Modeling for Accelerated Research in Transportation **SMART 2.0, Vehicle Technologies Office**, US Department of Energy (DOE) @ National Renewal Energy Laboratory, Denver CO
- Feasibility Analysis of Electric Roadways Through Localized Traffic, Cost, Adoption, and Environmental Impact Modeling, **ARPA-E, DOE**
- Investigating the Feasibility of Introducing Alternative Fuel Vehicles into Maintenance Fleet, Utah Department of Transportation (**UDOT**)
- Enhancement in 4-step Travel Demand Modeling considering individuals with disabilities incorporating physical community and environmental context*,* Funded by National Institute on Disability, Independent Living, and Rehabilitation Research (**NiDILRR**)

TEACHING EXPERIENCE

- **Teaching Assistant**, University at Buffalo; Utah State University, USA: 2021- present
  - Taught, provided feedback & graded students on assignments in class and in office hours
- **Assistant professor,** Pokhara University, Nepal: 2020-2021
  - Project planning & Control, Graduate Course – achieved 100% pass rate for MSc 1st semester
- **Lecturer,** Tribhuvan University, Nepal: 2013-2014
  - Transportation Engineering, Undergraduate Course – improved pass rates than in previous year
  - Supervised 2 Undergraduate Final projects on transportation planning and design

SKILLS

- **Programming and Optimization:** Python, R, GAMS, GurobiPy, CPLEX
- **Mapping and Visualization Software:** ArcGIS Pro, QGIS, ArcPy, matplotlib, seaborn
- **Machine/Deep Learning Frameworks**: SVM, SVD, DQN, Pytorch, HPC, PPO, MAPPO, MADDPG
- **Transportation Demand Modeling:** CUBE, Synchro, TransCAD, VISSIM, SmartRoad
- **Project management software:** Microsoft Project
- **Others:** Jupyter**,** Pandas, NumPy, scikit-learn, API web scraping, NLP, SQL

CERTIFICATIONS & LICENCES

- Engineer In Training (EIT), NCEES ID:2387261
- General Engineer, Civil, Nepal Engineering Council (NEC), since 2015

PROFESSIONAL AFFILIATIONS

- Member, TRB Standing Committee on Transit Operations (AP014), 08/2025-04/2028
- Member, TRB Standing Committee on Bus Transit Systems (AP050), 2025-2025
- Member, TRB Sub-committee on Freight Modeling (AT015(1)), 2023-2025
- Member, American Society of Civil Engineers (ASCE), 2025-2026
- Student member, Institute of Transportation Engineers (ITE), 2023-2025

LEADERSHIP ACTIVITIES

- Member, Tau Beta Pi, Utah Gamma Chapter, 2023-2025
- Member, Student Library Advisory Board, USU, 2023/24
- Utah State University Graduate Student Council (GSC), 2023/2024
- Vice-President, Civil Engineering Student Society-Nepal (CESS), 2012/2013, IOE Pulchowk, Nepal

AWARDS

- **Winner – Fellowship ($35,000)**, September 11th Memorial Program for Regional Transportation Planning; New York Metropolitan Transportation Council (NYMTC), 2025
- **Winner** - Institute of Transportation Engineers (ITE) Student Research Competition, Utah, 2024
- **Winner** - Prakash S Lad Scholarship for Courage and Perseverance in the College of Engineering (ENGR), Utah State University, 2023 **($2000.00)**
- Best student term project on data science titled "*Fake News Detection: Investigating ChatGPT."*
- **Dean list and University topper** in MSc Construction Management, Pokhara University, 2021

- **Outstanding MSc Student of the Year**, Pokhara University, 2019.

MEDIA MENTIONS

- Weitekamp, R. (2025, Oct). *Scholars Announced for NYMTC's 2025-26 9-11 Memorial Fellowship Program.* Media link here
- Maxwell, A. (2025, Sept 17). *CSEE student awarded September 11th Memorial fellowship for AI driven transportation solution for system efficiency.* Medial link here
- Dahle, S. (2024, Jan 23). *USU Engineering Scholar Showcases Transportation Research.* Media link here

REFERENCES

Ziqi Song, PhD Associate Professor Dept. of Civil, Str. and Env. Eng. University at Buffalo, SUNY Email: zqsong@buffalo.edu Phone: +1-716-645-3393	**Keith Christensen, PhD** Professor & Department Head LAEP Utah State University, Utah Email: keith.christensen@usu.edu Phone: +1-435-797-0507	**Jason Lustbador** Group Manager AVCI, CIMS National Laboratory of the Rockies Email: jason.lustbader@nrel.gov

PEER-REVIEWED CONFERENCE PRESENTATION

17. KC, M. B., Song, Z., Park, K., Chamberlin, B. & Christensen, K. (2025). Household Residential Location Choice Modeling for People with Disabilities in Utah. *ASCE International Conference on Transportation* Development, June 6-8, 2025, Arizona, USA (Accepted)

16. KC, M. B., & Song, Z. (2024). Multi-Stage, Multi-School Electric Bus Routing and Scheduling Optimization Using Metaheuristics. *Transportation Research Board (TRB) Annual Meeting, January 5-9, 2025*, Washington DC

15. KC, M. B., & Song, Z. (2024**)**. Integrated Optimized Charging and Economic Analysis of V2G Enabled School Bus System. *Transportation Research Board (TRB) Annual Meeting, January 5-9, 2025*, Washington DC

14. KC, Megh and Song, Ziqi (2024), " Inclusive Transport Policy: Mode Choice Behavior of People with Disabilities in Utah," *2024 ITE Mountain District Annual Meeting, Big Sky, Montana*, 19-21 June 2024 (Accepted)

13. KC, Megh; Song, Ziqi; Park, Keunhyun and Keith Christinsen (2024), "Inclusive Transportation System and Understanding Mode Choice Behavior of People with Disabilities: A Case Study in Utah", *6-Minute Showcase for the 2024 TRB Annual Meeting as part of the Young Member Council - Sustainability and* Resilience, Washington DC (awarded as recognized presentation)

12. KC, Megh and Song, Ziqi (2024), "Evolving School Transportation: A Comprehensive Approach to Bus Electrification with Dynamic Route Optimization and Partial Charging for Mixed Fleets", *Transportation Research Board (TRB) Annual Meeting, January 7-11, 2024*, Washington DC

11. KC, Megh; Song, Ziqi; Park, Keunhyun and Keith Christinsen (2024), "Understanding Mode Choice Behavior of People with Disabilities: A Case Study in Utah", *Transportation Research Board (TRB)* Annual Meeting, January 7-11, 2024, Washington DC

10. **KC, Megh** and Song, Ziqi (2023), "School Transportation Electrification: Simultaneous Route Optimization, Timetabling, and Partial Charging for Mixed Fleet," *2023 ITE Mountain District Annual Meeting, St. George, Utah*, 21-23 June 2023

9. KC, Megh and Song, Ziqi (2023), "Electric School Buses: Optimal Routing and Partial Charging," *USU* Student Research Symposium 2023, Logan, 12 April 2023

8. KC, Megh and Song, Ziqi (2023), "School Bus Electrification: Route Optimization, Timetabling, and Partial Charging," *College of Engineering Research Week, USU, Logan,* 4 April 2023.

RESEARCH OUTREACH ACTIVITIES

11. Reviewer: *Transportation Research Board Annual meeting, 2026*

10. Reviewer: *International Journal of Transportation Engineering and Technology (IJTET)*
9. Reviewer: *Journal of Transportation System and Engineering (JoTSE)*
8. Reviewer: *Energy 360*
7. Reviewer: *Transportation Research Record, Journal of Transportation Research Board*
6. Reviewer: *Transportation Research Board Annual meeting, 2025*
5. Reviewer: *Transportation Research Board Annual meeting, 2024*
4. Reviewer: *Transportation Research Board Annual meeting, 2023*
3. Reviewer: *Transportation Research Board Annual meeting, 2022*
2. Reviewer: *East African Scholars Journal of Economics, Business and Management*
1. Reviewer: *LEC Web-Based First International Conference on Engineering and Technology*